

# **G.A.T.S.**

# **User Manual**

*Version 2.0.0*

***GreekSoft Technologies Pvt. Ltd.***

*Siddivinayak Arcade, 3<sup>rd</sup> floor,*

*Akruli Cross Road No. 1,*

*Kandivali (E),*

*Mumbai*

**Proprietary Notice**

This document contains confidential information of GreekSoft Technologies Pvt. Ltd., which is provided for the sole purpose of permitting the recipient to application help document submitted herewith. In consideration of receipt of this document, the recipient agrees to maintain such information in confidence and to not reproduce or otherwise disclose this information to any person outside the group directly responsible for evaluation of its contents, except that there is no obligation to maintain the confidentiality of any information which was known to the recipient prior to receipt of such information from GreekSoft Technologies Pvt. Ltd., or becomes publicly known through no fault of recipient, or is received without obligation of confidentiality from a third party owing no obligation of confidentiality to GreekSoft Technologies Pvt. Ltd.

## **INDEX**

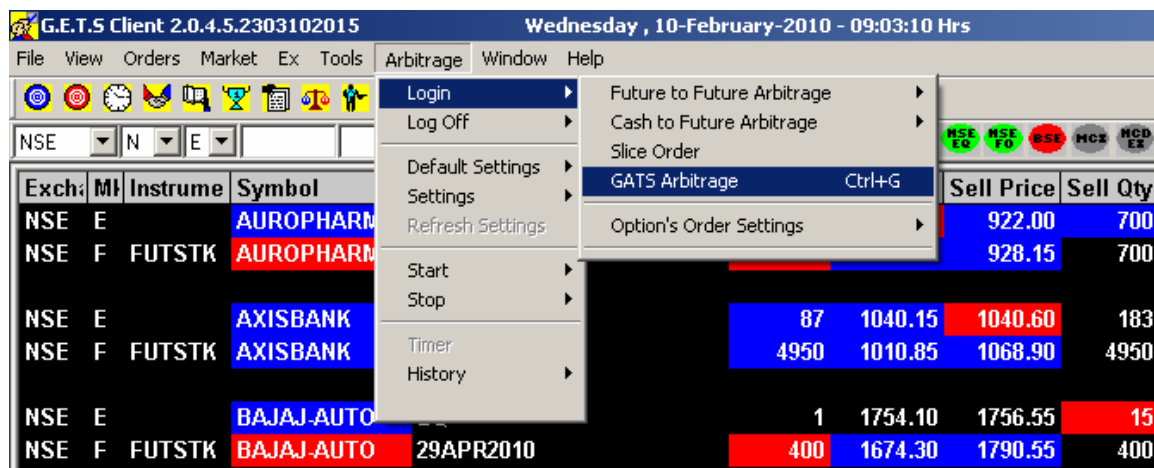
- I. Login*
- II. Input Parameters*
- III. Controlling Buttons*
- IV. GATS Default Settings*
- V. Execution Logic*
- VI. Order and Trade Book*
- VII. Net Position*

**Greeksoft Automated Trading System (GATS)** is used for gaining the advantage of volatile pricings between two different assets of cash or future market segments. The advantage of GATS is to reach the more accurate spread expected and faster execution in capturing the opportunity. Various arbitrage strategies those supported by GATS include:

- GATS Cash to Future Quote
- GATS Future to Future Opportunity
- GATS Future to Future Quote
- Pairs Trading Quote

### I. Login:

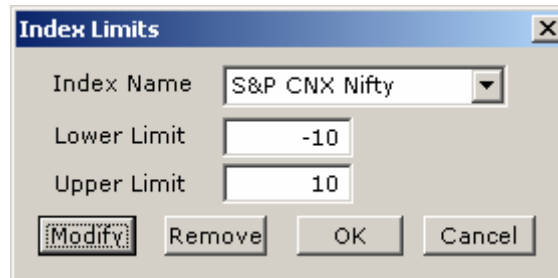
The user needs to login through GETS CTCL client, after which he will get the access of GATS Arbitrage from Arbitrage menu as shown below. CTRL+G from Market watch.



If the user enters wrong password for more than 3 times, his login will be blocked. To unblock, he has to request the system administrator.

During the login sequence to GATS arbitrage quote model, following checks must be defined, which will be validated in the post-login and execution.

**Index Limits Check** – When index crosses the set volatility range, then automated arbitrage will be stopped. Need to define the index and volatility range.



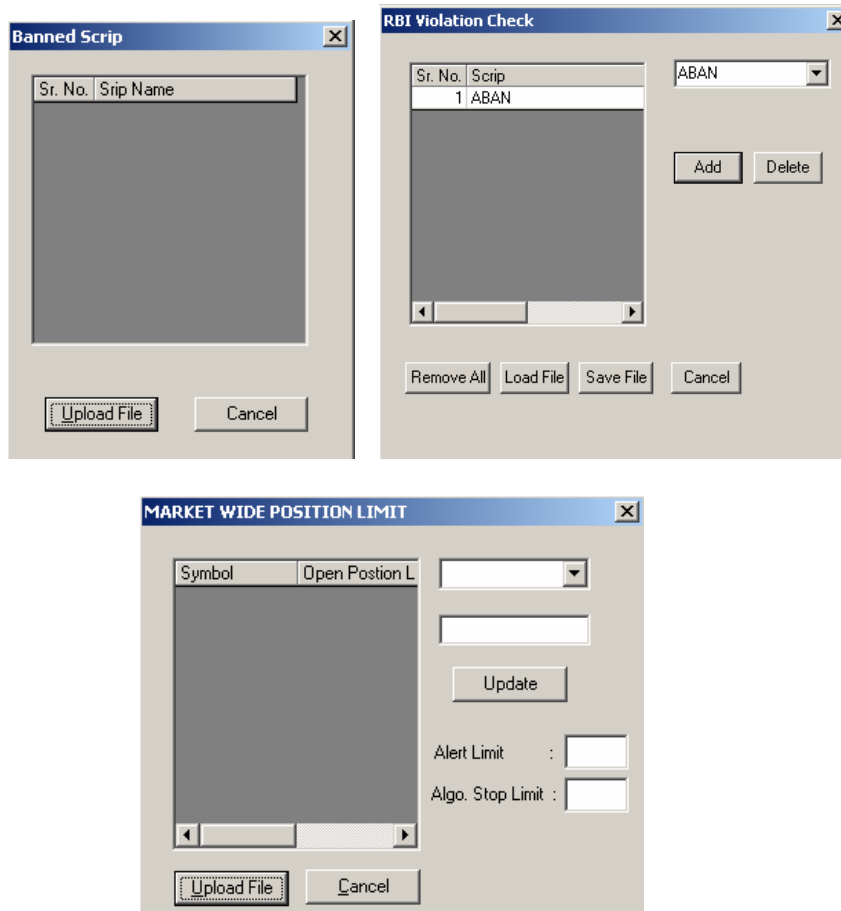
The 'Index Limits' dialog box contains the following fields and controls:

- Index Name:** A dropdown menu currently showing 'S&P CNX Nifty'.
- Lower Limit:** A text input field containing '-10'.
- Upper Limit:** A text input field containing '10'.
- Buttons:** 'Modify' (highlighted with a dashed border), 'Remove', 'OK', and 'Cancel'.

**Banned Scrip Check** – Trading in any banned scrip can be ruled out by uploading the banned scrip list. All violated symbols won't be loaded in the arbitrage settings form.

**RBI Violation Check** – Trading in any RBI violated scrip can be ruled out by uploading RBI violation list. All violated symbols won't be loaded in the arbitrage settings form.

**Market Wide Position Limit Check** – Trading in contracts, where market wide position limit reached, will be ruled out by uploading the market wide position limit list. All violated symbols won't be loaded in the arbitrage settings form.



The image displays three separate dialog boxes for managing trading restrictions:

- Banned Scrip:** Features a table with columns 'Sr. No.' and 'Scrip Name'. Below the table is an 'Upload File' button (highlighted with a dashed border) and a 'Cancel' button.
- RBI Violation Check:** Features a table with columns 'Sr. No.' and 'Scrip', containing one entry '1 ABAN'. To the right is a dropdown menu set to 'ABAN' and 'Add'/'Delete' buttons. Below the table are 'Remove All', 'Load File', 'Save File', and 'Cancel' buttons.
- MARKET WIDE POSITION LIMIT:** Features a table with columns 'Symbol' and 'Open Postion L'. To the right are two empty text input fields, an 'Update' button, and 'Alert Limit'/'Algo. Stop Limit' labels with corresponding input fields. At the bottom are an 'Upload File' button (highlighted with a dashed border) and a 'Cancel' button.

## II. Input parameters:

After successful login, the user will be prompted with the GATS dialog as shown below. He can call the dialog from market watch using short key CTRL+G.

The GATS dialog box is divided into several sections:

- First Order Section:** Contains fields for Exchange (NSE), Inst Type (EQUITY), Symbol (UNITECH), Expiry, Lot Size (1 N), Ord Ratio (4500), Pro/Cli (PRO), Account ID (07782), and Participant ID.
- Second Order Section:** Similar fields but with Inst Type set to FUTSTK and Expiry set to 29APR2010.
- Order Entry Section:** Includes checkboxes for 'BUY NSE UNITECH' and 'SELL NSE UNITECH', and a 'Buy First :: Sell Second' section with fields for Difference, Ord Lots, Tot Lots, Trd Lots, and Rem Lots.
- General Setting Section:** Includes radio buttons for 'Broadcast' and 'Timer', and fields for 'Interval', 'Best Buyer/Seller', 'Timer', 'Mkt Ord Upto', and 'Depth Upto'.
- Second Order Options Section:** Includes radio buttons for 'Market', 'Best Bid/Ask', 'Best Buyer/Seller', and 'Actual Diff'.
- Buttons:** Add, Modify, Remove, Remove All, Save File, Load File, Start All, Stop All, and Cancel.
- Table:** A table with columns: Rule, B Exchange, B Instrument, B Symbol, B Expiry, S Symbol, S Expiry, Price Diff, Price Diff Upper, Ordered Lot, Total Lot, Traded Lot, Base Order, Order Type, and S E. It contains two rows of data for UNITECH.

All input parameters are explained below.

1. **Exchange:** NSE
2. **Instrument Type:** EQUITY / FUTIDX / FUTSTK
3. **Symbol:** Symbols listed on NSE with respect to instrument selected
4. **Expiry:** Derivative expiry date
5. **Lot Size:** Displays the board lot of selected symbol
6. **Ord Ratio:** Represents the order quantity ratio based on the arbitrage type
7. **PRO/CLI:** Choose for proprietary trading / client trading
8. **Account ID:** Select the client account ID
9. **Participant ID:** NSE participant ID, in case of institutional client

Settings for Buy First & Sell Second Order

- 10. **Difference:** Input required absolute spread range
- 11. **Order Lots:** Input the no. of lots that has to be placed for each order
- 12. **Total Lots:** Input the total no. of lots to execute
- 13. **Trd Lots:** Displays the traded quantity for selected symbol
- 14. **Rem Lots:** Displays the remaining quantity for selected symbol

Settings for Sell First & Buy Second Order

- 15. **Difference:** Input required absolute spread range
- 16. **Order Lots:** Input the no. of lots that has to be placed for each order
- 17. **Total Lots:** Input the total no. of lots to execute
- 18. **Trd Lots:** Displays the traded quantity for selected symbol
- 19. **Rem Lots:** Displays the remaining quantity for selected symbol

Base Order Selection

- 20. **Base Order:** Options – Buy / Sell – Select any one of these

*Buy* – On selecting this option, after buy order gets fully traded, sell order will be sent to exchange.

*Sell* – On selecting this option, after sell order gets fully traded, buy order will be sent to exchange.

User can add same rule buy side as well as sell side for the same spread range or different spread range.

- 21. **Order Type:** NORMAL / 2LEG

*NORMAL (For Quote)* - On selecting this option, base order will be generated and upon receiving the trades of base order, against order will be generated.

*2LEG (For Opportunity)* - On selecting this option, 2LEG order will be generated, whenever the opportunity exists in the market.

### Quantity conditions

22. **Th Qty %** – Input the threshold quantity to be considered for base order generation, which is required in validating the quantity (percentage to lot size) availability in the market depth.
23. **Disc. Qty %** - Input the disclosed quantity percentage, which is considered for counter cash orders. If the against order is cash order, then disclosed quantity is not considered.
24. **Abs Qty %** - Input the absolute quantity percentage. If the absolute quantity percentage of the pending quantity of counter order gets traded, then against order will be submitted. If the counter order is of future, then this check is not considered.

### General Settings

25. **Broadcast / Timer Interval** – Input the pending against order modification method.

*Broadcast* – Modify the pending against order upon change in the market data

*Timer Interval* – Modify the pending against order upon expiration of the input time interval.

26. **Best Buyer / Seller** – Input the no. of modifications for the pending against order.
27. **Timer** – Input the time interval to modify the pending against order to market.
28. **Mkt Ord Upto** – Input the maximum price tick by which the pending order price will be modified. If this is defined as zero, pending order will be modified as market order.
29. **Depth Upto** – Input the market picture depth size consideration based on weighted average price calculation for threshold quantity check. Value must be in the range of 1 to 5.

### Opportunity Check for Base Order Generation

30. **Difference Limit** – Input the opportunity difference that must present in market, upon which the base order will be generated.
31. **Checking Time** – Input the time interval to check the opportunity difference limit in market.

### Second Order Options

#### 32. Market / Best Bid / Ask / Best Buyer / Seller / Actual Diff

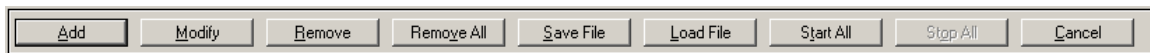
**Market** – Submit the against order as market order with price as zero, upon receiving the trades of first order

**Best Bid / Ask** – Submit the against order at best bid / ask price, upon receiving the trades of first order

**Best Buyer / Seller** – Submit the against order at best buyer / seller price, upon receiving the trades of first order

**Actual diff** – Submit against order at the actual difference set by the user, considering the avg. trade price of counter order.

### III. Controlling buttons:



*Add:* Adds the new arbitrage opportunity setting to the arbitrage farm

*Modify:* Modifies any previously added settings to the arbitrage farm

*Remove:* Removes any previously added settings from the arbitrage farm

*Remove All:* Removes all arbitrage settings from the arbitrage farm

*Save File:* Saves all the arbitrage settings present in the arbitrage farm

*Load File:* Loads previously saved arbitrage settings to arbitrage farm

*Start All:* Starts calculating the opportunity and submission of the orders

*Stop All:* Stops calculating the opportunity. Calling this will cancel all pending orders from order book operated by the arbitrage module

*Cancel:* Closes the GATS dialog

#### **IV. GATS Default Settings:**

After login to GATS, the user must have to set the default settings. The dialog can be called from Arbitrage -> Default Settings -> GATS Settings.

**GATS Default Settings**

Arbitrage Type: **Cash To Future**

Market Order Upto: **0.00** Rs. Absolute Qty Value: **100** %  
Best Ask/Bid: **2** times Threshold Qty: **100** %  
Timer: **5** secs Disclosed Qty: **0** %  
Depth Upto: **5** Near Expiry: **29APR2010**  
Far Expiry:

Order Priority:  
☐ Buy ☐ Sell ☐ Both ☒ Default

Conditional (Applicable For GATS)  
☒ Allow Pair Trading ☒ IOC Trade Popup

Default Option:  
☐ Cash Future ☐ Future to Future ☐ Future to Future Opp ☒ Default

**Apply** **Cancel**

- Arbitrage Type:** Cash to Future / Future to Future
- Market Order Upto:** Define market order upto value; must be multiple of 0.05
- Timer:** Define timer; after expiration of this timer pending order gets modified to market
- Depth Upto:** Define market depth; value must be in the range of 1 to 5
- Absolute Qty %:** Define absolute quantity %; in case of cash to future
- Threshold Qty %:** Define threshold quantity % for base order generation
- Disclosed Qty %:** Define disclose quantity %; in case of cash to future
- Near Expiry:** Define expiry for cash to future  
Define first expiry for future to future
- Far Expiry:** Define second expiry for future to future
- Order Priority:** Select Buy / Sell / Default; for base order generation
- Allow Pair Trading:** Check to allow pair trading
- IOC Trade Popup:** Check for message popup upon total lots completion
- Default Option:** Default strategy when called CTRL+G; if the default option is selected, then calling CTRL+G from market watch by selecting EQ scrip will open GATS cash to future; if Future scrip is selected then GATS future to future will be opened

## **V. Execution Logic:**

Execution logic is divided into below cases based on the type of arbitrage orders and base order option.

### **Order Type: NORMAL (for Quote)**

1. Buy First and Sell Second with base order option as Buy
2. Buy First and Sell Second with base order option as Sell
3. Sell First and Buy Second with base order option as Sell
4. Sell First and Buy Second with base order option as Buy

### **Order Type: 2LEG (for Opportunity)**

5. 2LEG order with base price as Buy
6. 2LEG order with base price as Sell

Various arbitrage strategies that can be executed are

Cash to Future Quote

- Buy Cash :: Sell Future
- Sell Cash :: Buy Future

Future to Future Quote

- Buy Expiry1 :: Sell Expiry2

Future to Future Opportunity

Pairs Trading Quote

- Buy Symbol1 :: Sell Symbol2

All cases are explained below.

**Order Type: NORMAL (for Quote)**

**Case 1: Buy First and Sell Second with first order base option as Buy**

**Strategy used for explanation:** Buy cash and Sell future with base order as Buy

After filling the input parameters, when the user call the Start control button, first buy cash counter order will be placed on satisfying the below conditions.

In case of single order lot,

Remaining quantity – should be more than zero  
Opportunity price difference – must available in market

In case of multiple order lots,

Remaining quantity – should be more than zero  
Opportunity price difference – must available in market and  
Depth upto & Threshold quantity – Total future bid quantity, considering the set market depth level, should be more than or equal to set % of threshold quantity

And the order parameters will be as follows.

Order Quantity – Set order lots considering the remaining quantity  
Order Price – Current Future Best Bid minus difference set  
Disclosed quantity – Set % of disclosed quantity

**First Order Cases**

***Counter cash order is pending:***

*Conditions checked:* Difference set, Market Depth

For every tick change in the future best bid price, the price of pending cash buy order will be modified, considering the set difference.

In case of multiple lots, if the pending order quantity not available in the market depth on futures bid side, then the cash order will be canceled and fresh counter cash order will be placed based on the total future bid quantity & market depth.

Modified cash order price - Current Future Best Bid minus difference set

***Counter cash order is partially traded:***

*Conditions checked:* Absolute quantity & Timer

If traded quantity is below absolute quantity % till the set time interval – Cancel the pending buy cash order and traded cash quantity will be reversed at market price.

In case of multiple order lots, pending buy order quantity will be modified to the nearest lot size on higher side and order price to market. And places the future sell order with cash side traded quantity as order quantity.

If executed quantity is above absolute quantity % - Future order will be generated

***Pending counter buy cash order after against sell future order is placed:***

Modify the pending cash order to market along with sell future order generation.

***Counter cash order is fully traded:***

Future order will be generated considering the below parameters.

Order Quantity – traded quantity counter buy cash order

Order Price (if Second order actual difference checked) – Avg. trade price of cash order executed + Set difference

Order Price (if Second order actual difference not checked) – Order price of cash order executed + Set difference

**Second Order Cases**

***Against future order is pending / partially traded:***

Conditions checked: Pending order modification type (broadcast / timer), Best Buyer / Seller modifications, Market Order Upto, Time interval

If the pending order modification type is set as broadcast, then the pending future order will be modified upon every new tick of market data. If the pending order modification type is set as timer, then the pending future order will be modified after the set time interval.

Pending future sell order will be modified to best ask price for the no. of best buyer/seller modifications mentioned after the set time interval. If the order is pending even after the no. of modifications, then the pending future order will be modified as below.

If Market Order Upto value is zero - modify the pending future order as market order.  
If Market Order Upto value is other than zero - modify the pending future sell order price by subtracting the set market order upto value from pending order price. The same process will be repeated for 10 times and then it will hit to market.

***Against future order is fully traded:***

Order cycle completed. If remaining quantity more than zero, Case 1 will be repeated.

**Case 2: Buy First :: Sell Second with base order option as Sell**

**Strategy used for explanation:** Buy cash and Sell future with base order as Sell

After filling the input parameters, when the user call the Start control button first sell future counter order will be placed on satisfying the below conditions (for single / multiple order lots).

Remaining quantity – should be more than zero

Opportunity price difference – must available in market

Depth upto & Threshold quantity – Total cash ask quantity, considering the set market depth level, should be more than or equal to set % of threshold quantity

And the counter future order parameters will be as follows.

Order Quantity – Set order lots considering the remaining quantity

Order Price – Current Cash Best Ask plus difference set

**First Order Cases**

*Counter future order is pending*

*Conditions checked:* Difference set, Market Depth

For every tick change in the cash best ask price, the price of pending future sell order will be modified, considering the set difference.

In case of multiple lots, if the pending order quantity is not available in the market depth on cash ask side, then the future order will be canceled and fresh counter future order will be placed based on the total cash ask quantity & market depth.

Modified future order price - Current Cash Best Ask plus difference set

*Counter future order is partially traded (in case of multiple lots)*

Cancel the pending future sell order and place against cash buy order.

*Counter future order is fully traded*

Against cash buy order will be generated considering the below parameters.

Order Quantity – traded quantity of counter sell future order

Order Price (if Second order actual difference checked) – Avg. trade price of future sell order executed – Set difference

Order Price (if Second order actual difference not checked) – Order price of future sell order executed – Set difference

### **Second Order Cases**

*Against cash order is pending / partially traded*

Conditions checked: Pending order modification type (broadcast / timer), Best Bid/Ask modifications, Market Order Upto, Time interval

If the pending order modification type is set as broadcast, then the pending cash order will be modified upon every new tick of market data. If the pending order modification type is set as timer, then the pending cash order will be modified after the set time interval.

Pending cash buy order will be modified to best bid price for the no. of best buyer/seller modifications mentioned after the set time interval. If the order is pending even after the no. of modifications, then the pending cash order will be modified as below.

If Market Order Upto value is zero - modify the pending cash order as market order.

If Market Order Upto value is other than zero - modify the pending cash buy order price by adding the set market order upto value from pending order price. The same process will be repeated for 10 times and then it will hit to market.

*Against cash order is fully traded*

Order cycle completed. If remaining quantity more than zero, Case 2 will be repeated.

**Case 3: Sell First :: Buy Second with base order option as Sell**

**Strategy used for explanation:** Sell cash and Buy future with base order as Sell

After filling the input parameters, when the user call the Start control button, first sell cash counter order will be placed on satisfying the below conditions.

In case of single order lot,

Remaining quantity – should be more than zero  
Opportunity price difference – must available in market

In case of multiple order lots,

Remaining quantity – should be more than zero  
Opportunity price difference – must available in market and  
Depth upto & Threshold quantity – Total future ask quantity, considering the set market depth level, should be more than or equal to set % of threshold quantity

And the order parameters will be as follows.

Order Quantity – Set order lots considering the remaining quantity  
Order Price – Current Future Best Ask plus difference set  
Disclosed quantity – Set % of disclosed quantity

**First Order Cases**

***Counter cash order is pending:***

Conditions checked: Difference set, Market depth

For every tick change in the future best ask price, the price of pending cash sell order will be modified, considering the set difference.

In case of multiple lots, if the pending order quantity not available in the market depth on futures ask side, then the cash order will be canceled and fresh counter cash order will be placed based on the total future ask quantity & market depth.

Modified cash order price - Current Future Best Ask plus difference set

***Counter cash order is partially traded:***

Conditions checked: Absolute quantity & Timer

If traded quantity is below absolute quantity % till the set time interval – Cancel the pending sell cash order and traded cash quantity will be reversed at market price.

In case of multiple order lots, pending sell order quantity will be modified to the nearest lot size on higher side and order price to market. And places the future buy order with cash side traded quantity as order quantity.

If executed quantity is above absolute quantity % - Future order will be generated

***Pending counter sell cash order after against buy future order is placed:***

Modify the pending cash order to market along with the future order generation.

***Counter cash order is fully traded:***

Future order will be generated considering the below parameters.

Order Quantity – traded quantity counter sell cash order

Order Price (if Second order actual difference checked) – Avg. trade price of cash order executed – Set difference

Order Price (if Second order actual difference not checked) – Order price of cash order executed – Set difference

**Second Order Cases**

***Against future order is pending / partially traded:***

Conditions checked: Pending order modification type (broadcast / timer), Best Bid/Ask modifications, Market Order Upto, Time interval

If the pending order modification type is set as broadcast, then the pending future order will be modified upon every new tick of market data. If the pending order modification type is set as timer, then the pending future order will be modified after the set time interval.

Pending future buy order will be modified to best bid price for the no. of best buyer/seller modifications mentioned after the set time interval. If the order is pending even after the no. of modifications, then the pending future order will be modified as below.

If Market Order Upto value is zero - modify the pending future order as market order.  
If Market Order Upto value is other than zero - modify the pending future buy order price by adding the set market order upto value from pending order price. The same process will be repeated for 10 times and then it will hit to market.

***Against future order is fully traded:***

Order cycle completed. If remaining quantity more than zero, Case 3 will be repeated.

**Case 4: Sell First :: Buy Second with base order option as Buy**

**Strategy used for explanation:** Sell cash and Buy future with base order as Buy

After filling the input parameters, when the user call the Start control button first buy future counter order will be placed on satisfying the below conditions (for single / multiple order lots).

Remaining quantity – should be more than zero

Opportunity price difference – must available in market

Depth upto & Threshold quantity – Total cash bid quantity, considering the set market depth level, should be more than or equal to set % of threshold quantity

And the counter future order parameters will be as follows.

Order Quantity – Set order lots considering the remaining quantity

Order Price – Current Cash Best Bid minus difference set

**First Order Cases**

*Counter future order is pending*

Conditions checked: Difference set, Market depth

For every tick change in the cash best bid price, the price of pending future buy order will be modified, considering the set difference.

In case of multiple lots, if the pending order quantity is not available in the market depth on cash bid side, then the future order will be canceled and fresh counter future order will be placed based on the total cash bid quantity & market depth.

Modified future order price - Current Cash Best Bid minus difference set

*Counter future order is partially traded (in case of multiple lots)*

Cancel the pending future buy order and place against cash sell order.

*Counter future order is fully traded*

Against cash sell order will be generated considering the below parameters.

Order Quantity – traded quantity of counter buy future order

Order Price (if Second order actual difference checked) – Avg. trade price of future buy order executed + Set difference

Order Price (if Second order actual difference not checked) – Order price of future buy order executed + Set difference

### **Second Order Cases**

*Against cash order is pending / partially traded*

Conditions checked: Pending order modification type (broadcast / timer), Best Bid/Ask modifications, Market Order Upto, Time interval

If the pending order modification type is set as broadcast, then the pending cash order will be modified upon every new tick of market data. If the pending order modification type is set as timer, then the pending cash order will be modified after the set time interval.

Pending cash sell order will be modified to best ask price for the no. of best buyer/seller modifications mentioned after the set time interval. If the order is pending even after the no. of modifications, then the pending cash order will be modified as below.

If Market Order Upto value is zero - modify the pending cash order as market order.

If Market Order Upto value is other than zero - modify the pending cash sell order price by subtracting the set market order upto value from pending order price. The same process will be repeated for 10 times and then it will hit to market.

*Against cash order is fully traded*

Order cycle completed. If remaining quantity more than zero, Case 4 will be repeated.

**Order Type: 2LEG (for Opportunity)**

After filling the input parameters, the user must call the Start control button. This activates the application to calculate the live spread in market and compares with the set price difference. If the live spread is equal to or greater than the set spread, then a 2LEG order with order validity as IOC having two legs (first leg as far expiry and second leg as near expiry) will be placed for the available lots in the market.

The calculation of live spread is based on below formulae.

Buy Near :: Sell Far

Calculated live spread = (Best Bid of Far expiry – Best Ask of Near expiry)

Sell Near :: Buy Far

Calculated live spread = (Best Bid of Near expiry – Best Ask of Far expiry)

In case of multiple order lots, weighted average price is considered based on the market depth.

As the order validity is IOC, either both the legs gets executed or entire order gets cancelled. So there will be no risk of partial execution on any side, thus ruling out the price risk.

**Case 5: 2LEG order with base price as Buy**

If the base price is selected as Buy, then the order price of sell order will be derived using the buy order price and difference.

Order Price of Buy Order = Best Ask Price

Order Price of Sell Order = Order Price of Buy Order + Difference

**Case 6: 2LEG order with base price as Sell**

If the base price is selected as Sell, then the order price of buy order will be derived using the sell order price and difference.

Order Price of Sell Order = Best Bid Price

Order Price of Buy Order = Order Price of Sell Order – Difference

## VI. Order Book & Trade Book:

All the orders resulting from the execution of cash vs. future arbitrage quote model gets updated in pending order book (F3). And all trades get updated in previous trade book (F8).

| Pending Orders Of All Scrips                       |           |      |        |      |        |      |       |             |       |      |     |               |       |          |             |                  |           |                 |         |       |
|----------------------------------------------------|-----------|------|--------|------|--------|------|-------|-------------|-------|------|-----|---------------|-------|----------|-------------|------------------|-----------|-----------------|---------|-------|
| GreekC                                             | RefNo.    | Exch | Trader | Inst | Nal    | Code | Symb  | Ser/Exp/Grc | Strik | Op   | B/S | Pend          | Price | Total    | Prc         | Clie             | Client N. | Exch. Order No. | Settlor | Valid |
| AJAY1                                              | 200002808 | NSE  | 18192  | F    | FUTIDX | 8986 | NIFTY | 27AUG2009   |       | Buy  | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000122 | 07782     | DAY             |         |       |
| AJAY1                                              | 200002808 | NSE  | 18192  | F    | FUTIDX | 5999 | NIFTY | 24SEP2009   |       | Sell | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000122 | 07782     | DAY             |         |       |
| AJAY1                                              | 200002807 | NSE  | 18192  | F    | FUTIDX | 8986 | NIFTY | 27AUG2009   |       | Buy  | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000121 | 07782     | DAY             |         |       |
| AJAY1                                              | 200002807 | NSE  | 18192  | F    | FUTIDX | 5999 | NIFTY | 24SEP2009   |       | Sell | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000121 | 07782     | DAY             |         |       |
| AJAY1                                              | 200002806 | NSE  | 18192  | F    | FUTIDX | 8986 | NIFTY | 27AUG2009   |       | Buy  | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000120 | 07782     | DAY             |         |       |
| AJAY1                                              | 200002806 | NSE  | 18192  | F    | FUTIDX | 5999 | NIFTY | 24SEP2009   |       | Sell | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000120 | 07782     | DAY             |         |       |
| AJAY1                                              | 200002811 | NSE  | 18192  | F    | FUTIDX | 8986 | NIFTY | 27AUG2009   |       | Buy  | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000125 | 07782     | DAY             |         |       |
| AJAY1                                              | 200002811 | NSE  | 18192  | F    | FUTIDX | 5999 | NIFTY | 24SEP2009   |       | Sell | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000125 | 07782     | DAY             |         |       |
| AJAY1                                              | 200002810 | NSE  | 18192  | F    | FUTIDX | 8986 | NIFTY | 27AUG2009   |       | Buy  | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000124 | 07782     | DAY             |         |       |
| AJAY1                                              | 200002810 | NSE  | 18192  | F    | FUTIDX | 5999 | NIFTY | 24SEP2009   |       | Sell | 50  | 0.00          | 50    | CLI 0H01 | HITESH      | 2009080310000124 | 07782     | DAY             |         |       |
| G.E.T.S Pending Orders Of All Scrips NIFTY09AUGFUT |           |      |        |      |        |      |       |             |       |      |     | BuyQty : 3100 |       |          | SellQty : 0 |                  |           |                 |         |       |

| Trades-All                                                                                                         |       |         |   |        |          |            |             |         |     |      |     |         |     |                |        |          |        |           |
|--------------------------------------------------------------------------------------------------------------------|-------|---------|---|--------|----------|------------|-------------|---------|-----|------|-----|---------|-----|----------------|--------|----------|--------|-----------|
| Exch                                                                                                               | Membr | TraderC | N | Inst   | Nam      | Symbol/Sci | Ser/Exp/Gro | Strike  | Opt | B/S  | Qty | Price   | Pro | Exch. Order No | Trader | Time     | GreekC | Ref No    |
| NSE                                                                                                                | 07782 | 18192   | F | FUTIDX | NIFTY    | 24SEP2009  |             |         |     | Sell | 50  | 4691.70 | CLI | 09080410000044 | 20019  | 12:13:50 | AJAY1  | 100000071 |
| NSE                                                                                                                | 07782 | 18192   | F | FUTIDX | NIFTY    | 24SEP2009  |             |         |     | Sell | 50  | 4691.70 | CLI | 09080410000044 | 20018  | 12:13:50 | AJAY1  | 100000071 |
| NSE                                                                                                                | 07782 | 18192   | F | FUTIDX | NIFTY    | 24SEP2009  |             |         |     | Sell | 50  | 4691.70 | CLI | 09080410000044 | 20017  | 12:13:50 | AJAY1  | 100000071 |
| NSE                                                                                                                | 07782 | 18192   | F | FUTIDX | NIFTY    | 27AUG2009  |             |         |     | Buy  | 50  | 4689.00 | CLI | 09080410000040 | 20016  | 12:13:44 | AJAY1  | 100000067 |
| NSE                                                                                                                | 07782 | 18192   | F | FUTIDX | NIFTY    | 24SEP2009  |             |         |     | Sell | 50  | 4692.35 | CLI | 09080410000042 | 20015  | 12:13:42 | AJAY1  | 100000069 |
| NSE                                                                                                                | 07782 | 18192   | F | FUTIDX | NIFTY    | 27AUG2009  |             |         |     | Buy  | 50  | 4689.25 | CLI | 09080410000041 | 20014  | 12:13:42 | AJAY1  | 100000068 |
| NSE                                                                                                                | 07782 | 18192   | O | OPTIDX | NIFTY    | 27AUG2009  |             | 4600.00 | PE  | Sell | 50  | 128.30  | CLI | 09080410000039 | 20013  | 12:13:18 | AJAY1  | 100000066 |
| NSE                                                                                                                | 07782 | 18192   | O | OPTIDX | NIFTY    | 27AUG2009  |             | 4600.00 | CE  | Buy  | 50  | 214.90  | CLI | 09080410000038 | 20012  | 12:13:16 | AJAY1  | 100000065 |
| NSE                                                                                                                | 07782 | 18192   | F | FUTSTK | RELIANCE | 27AUG2009  |             |         |     | Buy  | 150 | 2043.70 | CLI | 09080410000037 | 20010  | 12:10:55 | AJAY1  | 100000064 |
| NSE                                                                                                                | 07782 | 18192   | F | FUTSTK | RELIANCE | 27AUG2009  |             |         |     | Sell | 150 | 2045.85 | CLI | 09080410000032 | 20004  | 12:10:14 | AJAY1  | 100000059 |
| NSE                                                                                                                | 07782 | 18192   | F | FUTSTK | RELIANCE | 27AUG2009  |             |         |     | Buy  | 150 | 2046.00 | CLI | 09080410000031 | 20003  | 12:10:04 | AJAY1  | 100000058 |
| G.E.T.S NIFTY09SEP2009 BuyQty : 0 BuyVal : 0.00 BuyATP : 0.00 SellQty : 550 SellVal : 2580467.60 SellATP : 4691.76 |       |         |   |        |          |            |             |         |     |      |     |         |     |                |        |          |        |           |

## VII. Net Positions:

All the position summary of previous trades updated in net position dialog (ALT+F6)

| Net Position            |              |        |    |                  |             |           |                     |    |    |             |           |         |             |            |          |                   |            |          |                |  |  |                   |  |  |             |  |  |
|-------------------------|--------------|--------|----|------------------|-------------|-----------|---------------------|----|----|-------------|-----------|---------|-------------|------------|----------|-------------------|------------|----------|----------------|--|--|-------------------|--|--|-------------|--|--|
| Exchange                | Mkt Type     | Symbol |    |                  | Periodicity | User      | Security / Contract |    |    |             |           |         |             |            |          |                   |            |          |                |  |  |                   |  |  |             |  |  |
| NSE                     | All          | All    |    |                  | Daily       | AJAY1     |                     |    |    |             |           |         |             |            |          |                   |            |          |                |  |  |                   |  |  |             |  |  |
| <a href="#">Refresh</a> |              |        |    |                  |             |           |                     |    |    |             |           |         |             |            |          |                   |            |          |                |  |  |                   |  |  |             |  |  |
| Excl                    | User         | Client | MI | Instrument       | Symbol      | Ser/Exp   | Strike              | P  | Op | Buy Qt      | Buy Val   | Buy Avg | Sell Qt     | Sell Val   | Sell Avg | Net               | Net Val    | MTM G/L  |                |  |  |                   |  |  |             |  |  |
| NSE                     | AJAY1        | 07782  | O  | OPTIDX           | NIFTY       | 27AUG2009 | 4600.00             | CE |    | 50          | 10745.00  | 214.90  | 0           | 0.00       | 0.00     | 50                | 10745.00   | 357.50   |                |  |  |                   |  |  |             |  |  |
| NSE                     | AJAY1        | 07782  | O  | OPTIDX           | NIFTY       | 27AUG2009 | 4600.00             | PE |    | 0           | 0.00      | 0.00    | 50          | 6415.00    | 128.30   | -50               | -6415.00   | 237.50   |                |  |  |                   |  |  |             |  |  |
| NSE                     | AJAY1        | 07782  | F  | FUTIDX           | NIFTY       | 27AUG2009 |                     |    |    | 100         | 468912.50 | 4689.13 | 0           | 0.00       | 0.00     | 100               | 468912.50  | 1257.00  |                |  |  |                   |  |  |             |  |  |
| NSE                     | AJAY1        | 07782  | F  | FUTIDX           | NIFTY       | 24SEP2009 |                     |    |    | 0           | 0.00      | 0.00    | 550         | 2580467.50 | 4691.76  | -550              | 2580467.50 | -7282.00 |                |  |  |                   |  |  |             |  |  |
| NSE                     | AJAY1        | 07782  | F  | FUTSTK           | RELIANCE    | 27AUG2009 |                     |    |    | 300         | 613455.00 | 2044.85 | 150         | 306877.50  | 2045.85  | 150               | 306577.50  | 532.50   |                |  |  |                   |  |  |             |  |  |
| Tota                    |              |        |    |                  |             |           |                     |    |    | 450         | 093112.50 |         | 750         | 2893760.00 |          | -300              | 1800647.50 | -4897.50 |                |  |  |                   |  |  |             |  |  |
|                         |              |        |    |                  |             |           |                     |    |    |             |           |         |             |            |          |                   |            |          |                |  |  |                   |  |  |             |  |  |
| G.E.T.S                 | Net Position |        |    | NIFTY09AUG4600CE |             |           | Buy Qty: 50         |    |    | Sell Qty: 0 |           |         | Net Qty: 50 |            |          | Buy Val: 10745.00 |            |          | Sell Val: 0.00 |  |  | Net Val: 10745.00 |  |  | MTM: 405.00 |  |  |

-----END OF DOCUMENT-----